

Real-Time Shape Recognition System

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Introduction

Shape recognition is an important task in image processing and pattern recognition.
Manual identification of shapes can be: Slow, Inconsistent and Prone to human error

The project develops a system that automatically detects and classifies geometric shapes.

The system works with:

- Live webcam input
- Static/uploaded images

Uses classical image processing techniques without pre-trained ML/DL models.

Problem statement

The Challenge: Manual inspection and sorting of geometric shapes in industrial, manufacturing, and educational settings are tedious, slow, and prone to human error.

The Solution: An automated computer vision tool that operates in real-time, providing consistent and precise classifications without the overhead of heavy deep-learning neural network architectures.

Objectives of the project

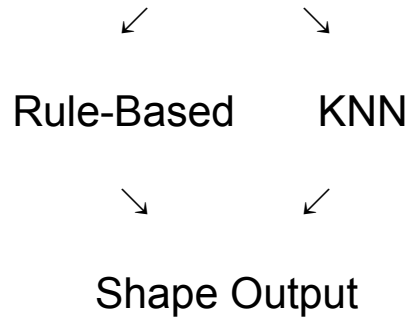
- Design a real-time image processing pipeline for shape detection.
- Extract meaningful geometric and statistical features.
- Implement two classification approaches: Rule-Based Classifier, K-Nearest Neighbours (KNN)
- Evaluate the system
- Develop an accessible application through: OpenCV desktop application, Streamlit web dashboard

Scope of the project

- Recognition of 7 geometric shapes
- Static image and live webcam input
- Multiple-shape detection
- Geometric feature extraction
- Rule-based and KNN classification
- Real-time OpenCV & Streamlit deployment
- Designed for clear, high-contrast shapes

System Methodology

Input Image / Webcam -> Grayscale Conversion -> Gaussian Blurring -> Adaptive Thresholding -> Morphological Closing -> Contour Detection -> Feature Extraction -> Classification



Feature Extraction Classification

Feature Extraction

- Vertex Count, Circularity → polygon structure and degree of circularity.
- Aspect Ratio, Extent, Solidity → dimensions, area coverage, and shape compactness.
- 7 Hu Moments → translation, scale, and rotation-invariant shape descriptors.
- Total: 12-dimensional feature vector, representing geometric structure rather than raw pixels.

Classification

- Rule-Based: uses circularity, vertex count, and aspect ratio to identify shapes.
- Examples: high circularity → Circle; 3 vertices → Triangle; 4 vertices + ratio ≈ 1 → Square; otherwise → Rectangle.
- KNN: $k = 5$, uses standardized features and the self-generated dataset.
- Trained from scratch with no pre-trained models, weights, or external dataset

Workflow

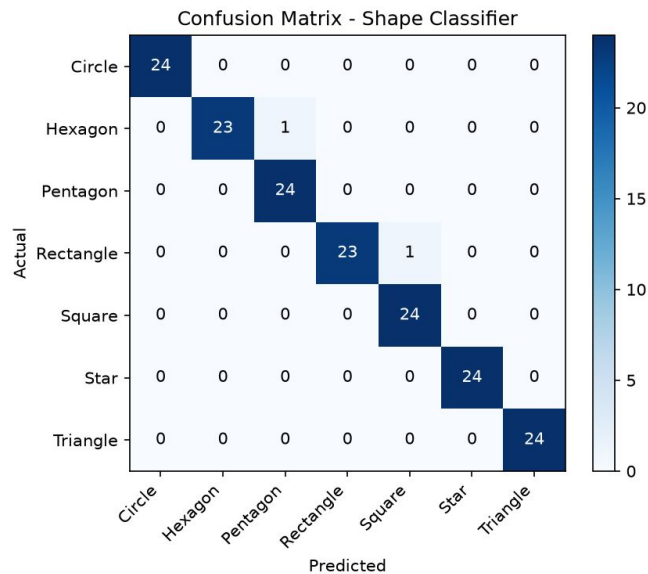
Input → Preprocessing → Contour Detection → Feature Extraction → Classification → Output

1. Input: Capture image from webcam or upload a static image.
2. Preprocessing: Convert to grayscale → Gaussian blur → adaptive thresholding → morphological closing.
3. Detection & Features: Detect contours → filter noise → extract 12-dimensional geometric features.
4. Classification: Apply Rule-Based / KNN classifier → identify the shape.
5. Output: Display shape label, bounding box, and number of detected shapes on the image/frame.

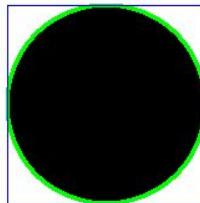
Implementation Techniques

Technology	Purpose
Python 3	Core programming
OpenCV	Image processing & contour detection
NumPy	Numerical computation
Scikit-learn	KNN, scaling & evaluation
Matplotlib	Performance visualization
Joblib	Model/scaler storage
Streamlit	Web dashboard

Result



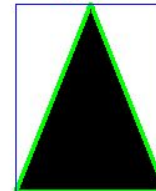
Circle (100%)



Square (60%)



Triangle (80%)



Rectangle (100%)



Metric	Score
Overall Accuracy	98.81%
Macro Precision	98.86%
Macro Recall	98.81%
Macro F1- score	98.81%

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Conclusion

- A complete real-time shape recognition system was successfully developed.
- Seven geometric shapes can be recognized.
- A self-generated dataset of 840 images was used.
- KNN achieved 98.81% test accuracy.
- The system was deployed through both:

THANK YOU